
Software Quality Assurance Management
Managing Software Organizations
Three-Level Hierarchy
Top Management: Strategy and Vision
Middle Management: Bridging Strategy and Operations
Project Management: Execution and Delivery
Management Roles
Executives: CEO, CTO
Department Managers
Project Managers and Team Leaders
Structure Variations
Organizational Size
Product vs Service focus
Geographic Distribution
Managing Software Quality
SQA Framework Foundations
Collective Contribution
Procedure Compliance
Professional Execution
SQA Participants
Managers: Execs, Dept heads, Project leads
Testers: Frontline defect detection
SQA Professionals: Trustees, Units, Committees
Involvement Levels
Full-Time: SQA Unit, Testing Department
Part-Time: Committees, Forums, Managers
Defect Prevention (CMM Level 5)
Core Objectives
Identify Root Causes
Incorporate Lessons Learned
Proactive Process Improvement
Key Practices
Defect Causal Analysis (DCA)
Action Proposals & Follow-Up
Preventive Action & Process Change
Organizational Learning
Management Involvement
Expected Benefits
Reduced Rework Costs
Improved Product Quality
Quantitative Enhancement
Culture of Quality
ISO Alignment
Corrective and Preventive Action (CAPA)
ISO 9000-3 / ISO/IEC 12207 Standards